Promastigote: This forms are found in the mid-gut of sand fly and in the culture media. The fully developed promastigotes are long, slender and spindle – shaped. They measure 15µm to 25µm in length and 1.5µm to 3.5µm in breadth. A single nucleus is situated at the centre. The kinetoplast lies near the anterior end. The flagellum is single, delicate and measures 15µm–28µm (Figure 8.8).

Infobits

There are 3 main forms of Leishmaniases – Visceral (also known as Kala-agar and the most serious form of the disease), cutaneous (the most common) and mucocutaneous. The disease affects some of the poorest people on earth, and is associated with malnutrition, population displacement, poor housing, a weak immune system and lack of financial resources. Leishmaniasis is linked to environmental changes such as deforestation, building of dams, irrigation schemes, and urbanization.

PKDL occurs in all areas endemic for L.donovani but is commonest in East Africa and on the Indian subcontinent, where up to 50% and 10% of patients with kala-azar, respectively develop the condition. The frequency is reported to be declining in India.

Cutaneous Leishmaniasis

The clinical spectrum of cutaneous leishmaniasis (oriental sore) is broad and may mimic that of other skin conditions, such as staphyloccal or streptococcal infection, mycobacterial ulcer, leprosy, fungal infection, cancer, sarcoidosis and tropical ulcer.

8.4.4 Life – Cycle of Leishmania donovani

Leishmania donovani completes its life cycle in two different hosts. The complete life cycle is given in Figure 8.9.

Host	Forms
Human and other mammals (Example: Dogs)	Amastigote
Sandfly of Genus *Phlebotomus*	Promastigote

Development in Human

The parasite is transmitted to human and other vertebrate hosts by the bite of blood sucking female sandfly. During the blood meal, the sandfly deposists promastigotes on surface of the skin. These promastigotes are immediately phagocytosed by fixed macrophages of the host, in which they are transformed into amastigotes. The amastigotes multiply by binary fission within the macrophages. As many as 50 to 200 amastigotes may be present inside the enlarged cell. These are called LD bodies. The rupture of cell releases amastigotes in large numbers which inturn are free to infect other cells. Free amastigotes are subsequently carried by circulation. These forms invade monocytes of the blood and macrophages of the spleen, liver, bone marrow, lymph nodes and other tissues of the reticuloendothelial cells.

Development in sandfly

Female sandfly during a blood meal ingest free, as well as intracellular amastigotes in the blood. In the mid gut of the sandfly, the amastigotes are transformed within 72 hours to flagellated promastigotes.

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